Bags, Vectors & Transformers

Your Turn: BERT on New Data · Day 3, Part 4

Denise J. Roth, PhD Candidate
Strategic Communication Group · Wageningen University & Research

Your Turn: BERT on New Data

Same methods as on the bill titles, on data that's harder in different ways. **Each table takes one dataset.**

Environmental claims

Does a company sentence make an environmental claim? Greenwashing research.

2,647 sentences · yes / no

Climate sentiment

Does a paragraph from a corporate report talk about climate risk or opportunity?

1,320 paragraphs · risk / neutral / opportunity

PUBHEALTH

Is a public health claim true? Can a model tell from the wording alone?

12,258 claims · true / false / mixture / unproven

YOUR TURN Open `2_bert_new_data_exercises` in the course repo (`day3/notebooks`), set `DATASET` at the top, place your bets, and work through steps 1 to 7. About **45 minutes**, then we compare.

Report Back

One or two sentences per table:

1. Which approach won, and by how much? Bigger or smaller gap than on the bills?
2. How did your bets do?
3. One error worth sharing: model wrong, or label wrong?
4. Would you use this classifier in a paper? What would you check first?

Test macro-F1

	TF-IDF	zero-shot	fine-tuned	our bet
Bill titles				
Env. claims				
Climate				
PUBHEALTH				

Where did BERT help most, and where did nothing really work? If a label depends on facts about the world rather than on the wording, no text classifier can learn it.

Errors Travel Into Your Analysis

A classifier is rarely the end goal. You use its labels to **count**, **compare groups**, **follow trends**, or as **variables in a regression**. Every error comes along.

Errors spread evenly

Mostly add noise: differences between groups look smaller than they are.

Errors that depend on the group

If the model works better for some parties, years or sources than others, it can create differences that aren't there, or hide ones that are.

What to do

- ✔ Look at **per-label** scores, and read the errors.
- ✔ Check error rates **separately for the groups you'll compare**.
- ✔ Compare true and predicted results on a labeled sample (Part B, Section 11.10).
- ✔ Correcting estimates for classification errors: bonus module 12.

What to Report in a Paper

When you use a transformer for measurement, readers need enough to judge your work and to reproduce it.

- **Model**: the exact name on the Hub (and version)
- **Training data**: source, size, inter-coder reliability, label distribution
- **Splits**: how, and that the test set was used once
- **Preprocessing**: maximum length, truncation, long documents
- **Settings**: learning rate, epochs, batch size, warmup, seed(s)
- **Results**: precision, recall and F1 per label; a confusion matrix
- **Baseline**: at least one simple model
- **Variance**: results across several seeds
- **Errors**: what kinds of texts the model gets wrong
- **Code and model**: shared, as far as your data allow

The same checklist is at the end of Part B of the notebook.

Day 3 in One Slide

- ✔ **Contextual embeddings**: a vector per word *per sentence*, so the "bank" problem goes away
- ✔ **Attention** lets every word use the whole sentence; BERT learns by **filling in blanks**
- ✔ Three ways to use a model: **off-the-shelf** or zero-shot, **frozen features**, **fine-tuning**
- ✔ On the Hub: **read the card**, validate on your own texts
- ✔ The **baseline** earns its place, the **labels** set the ceiling, and **errors** travel

Tomorrow

Sentence transformers and clustering, with Art Dewulf.

Large language models: what decoders add, annotating text with an LLM, open models, and distilling an LLM's labels into a fast BERT classifier.